

TASK 1 – REQUIREMENT & BUSINESS UNDERSTANDING

Mobile Phone Price Category Classification using Machine Learning

1. Business Problem

The company wants to develop a machine-learning solution that can classify mobile phones into different price categories based on their technical specifications. The system should learn the relationship between mobile-phone specifications and their corresponding price categories.

2. Project Objective

The main objective is to build a machine-learning model that takes the technical specifications of a mobile phone as input and predicts the appropriate price category.

3. Key Input Features

- RAM
- Internal storage
- Battery capacity
- Camera quality/specifications
- Screen resolution
- Processor-related features
- Connectivity features
- Other relevant hardware characteristics

4. Expected Output

The expected output of the system is the predicted price category of a mobile phone. The model should classify a phone into the appropriate category based on the input specifications.

5. Project Scope

- Understand the business requirement and expected outcome.
- Study the available mobile-phone dataset.
- Identify relevant features that can help predict price categories.
- Prepare the data for machine-learning analysis.
- Train and evaluate a suitable classification model.
- Use the trained model to predict the price category of new mobile-phone specifications.

6. Expected Business Benefit

The solution can help the company understand how mobile-phone specifications are associated with different price categories and can provide a consistent, data-driven method for classifying phones.

7. Task 1 Deliverable

Requirement Notes documenting the business problem, project objective, required input features, expected output, and overall project scope.

8. Task Summary

Item	Details
Task ID	TA-001
Task	Requirement & Business Understanding
Priority	High
Deliverable	Requirement Notes
Expected Outcome	Clear understanding of the ML business problem, inputs, outputs and scope

Prepared by: _____

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